

Problem 61 (Mediterranean MO 2006 Problem 1). We colour every point on the plane red or blue.

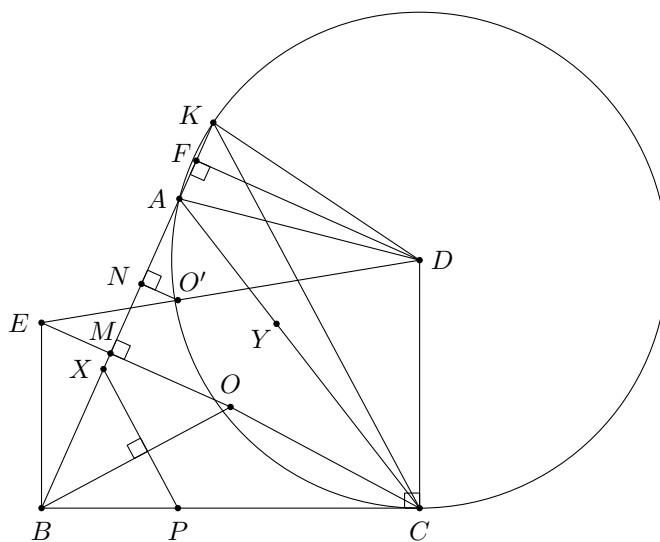
- (a) Is there a colouring such that there is exactly one red point on each unit circle?
(b) Is there a colouring such that there are exactly two red points on each unit circle?

Solution.

- (a) No. Suppose such a colouring exists. Clearly, there is a red point A . Also, there is a red point B lying on the unit circle with centre A . This implies $AB = 1$. Choose any point C such that $\triangle ABC$ is an equilateral triangle. Then the unit circle with centre C has at least two red points A and B , contradiction.
- (b) Yes. We colour every point on the Cartesian plane red if and only if its x -coordinate is an even integer. Note that every unit circle either meets exactly one vertical line $x = 2k$ where $k \in \mathbb{Z}$, or is tangent to two vertical lines of this form. In either case, the unit circle has exactly two red points on it.

□

Problem 62 (European Mathematical Cup 2015 Junior Problem 4). Let $\triangle ABC$ be an acute triangle with circumcentre O . Let D be the point on the perpendicular bisector of AC such that $DC \perp BC$. Let E be the point on the perpendicular bisector of AB such that $EB \perp BC$. Let P be an arbitrary point on the side BC . The line passing through P and perpendicular to BO meets the side AB at X . The line passing through P and perpendicular to CO meets the side AC at Y . Prove that the circumcentre of $\triangle AXY$ lies on the line DE .



Solution. Let M and N be the midpoints of AB and AX respectively. Let F be the projection of D onto the line AB , and let K be the symmetric point of A about F . Let O' be the intersection point of DE and the perpendicular bisector of AX .

Note that $DK = DA = DC$. So D is the centre of (ACK) . As $BC \perp CD$, BC is a tangent to (ACK) . Therefore, we have

$$\angle KCB = 180^\circ - \angle CAK = \angle BAC = 90^\circ - \angle CBO.$$

This proves $KC \perp BO$, and hence $KC \parallel XP$. By the intercept theorem, we have

$$\frac{BP}{BC} = \frac{BX}{BK} = \frac{2MN}{2MF} = \frac{MN}{MF} = \frac{EO'}{ED}.$$

This means $BP : PC = EO' : O'D$. By symmetry, DE meets the perpendicular bisector of AY at the same point O' . Thus, O' is the circumcentre of $\triangle AXY$, and we are done. □

Problem 63 (Bilkent University Problem of the Month 2018 July–August). Let x, y, z be positive real numbers satisfying $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} = 5$. Suppose $\sqrt{x}, \sqrt{y}, \sqrt{z}$ are the side lengths of a triangle. Prove that

$$\frac{x(y^2 - 2z^2)}{z} + \frac{y(z^2 - 2x^2)}{x} + \frac{z(x^2 - 2y^2)}{y} \geq 0.$$

Solution. We have

$$\left(\frac{x}{y} + \frac{y}{z} + \frac{z}{x}\right)(xy + yz + zx) = x^2 + y^2 + z^2 + xy + yz + zx + \frac{xy^2}{z} + \frac{yz^2}{x} + \frac{zx^2}{y}.$$

This implies

$$\frac{xy^2}{z} + \frac{yz^2}{x} + \frac{zx^2}{y} = 4(xy + yz + zx) - (x^2 + y^2 + z^2).$$

Thus,

$$\begin{aligned} & \frac{x(y^2 - 2z^2)}{z} + \frac{y(z^2 - 2x^2)}{x} + \frac{z(x^2 - 2y^2)}{y} \\ &= \frac{xy^2}{z} + \frac{yz^2}{x} + \frac{zx^2}{y} - 2(xy + yz + zx) \\ &= 2(xy + yz + zx) - (x^2 + y^2 + z^2). \end{aligned}$$

It remains to prove this is nonnegative. Indeed, by Heron's formula, the area of the triangle is

$$\sqrt{-(\sqrt{x})^4 - (\sqrt{y})^4 - (\sqrt{z})^4 + 2(\sqrt{x})^2(\sqrt{y})^2 + 2(\sqrt{y})^2(\sqrt{z})^2 + 2(\sqrt{z})^2(\sqrt{x})^2} = \sqrt{-x^2 - y^2 - z^2 + 2xy + 2yz + 2zx}.$$

So we must have $-x^2 - y^2 - z^2 + 2xy + 2yz + 2zx \geq 0$ as desired. $\square$

Problem 64 (Bilkent University Problem of the Month 2021 October). Let k be a positive integer. A positive integer n is *nice* if n is the sum of square of exactly k integers divisible by k . A positive integer m is *funny* if m is the sum of square of exactly k integers not divisible by k . Find all positive integers k such that every nice integer is funny.

Solution. The answer is any positive integer $k \neq 1, 2, 4$.

We say that k is *good* if every nice integer is funny. We first show that if k is good, so is any multiple of k . Let $m = kn$ and let $x_1, x_2, \dots, x_m$ be integers such that $m \mid x_i$ for all $1 \leq i \leq m$. Then since $k \mid x_i$ for all $1 \leq i \leq m$ and k is good, there exist integers $y_1, y_2, \dots, y_m$ such that

$$\sum_{i=k\ell+1}^{k(\ell+1)} x_i^2 = \sum_{i=k\ell+1}^{k(\ell+1)} y_i^2$$

for all $0 \leq \ell \leq n-1$ and $k \nmid y_i$ for all $1 \leq i \leq m$. Therefore we obtain

$$\sum_{i=1}^m x_i^2 = \sum_{i=1}^m y_i^2$$

and $m = kn \nmid y_i$ for all $1 \leq i \leq m$.

Next we show that all positive odd integers are good.

Lemma: Let k be a positive odd integer and let $x_1, x_2, \dots, x_k$ be integers with at least one of them not divisible by k . Then there exist integers $y_1, y_2, \dots, y_k$ such that none of them is divisible by k and

$$\sum_{i=1}^k (kx_i)^2 = \sum_{i=1}^k y_i^2.$$

Proof. Without loss of generality we may assume $k \nmid x_1$. Let $X = 2(x_1 + x_2 + \dots + x_k)$. If $k \mid X$, we replace x_1 by $-x_1$. As $k \nmid x_1$ and k is odd, $k \nmid 4x_1$ and hence we may assume $k \nmid X$. Then by the following identity

$$\sum_{i=1}^k (kx_i)^2 = \sum_{i=1}^k (X - kx_i)^2$$

letting $y_i = X - kx_i$ for all $1 \leq i \leq k$ works. $\square$

For a positive odd integer k , if a positive integer a is the sum of squares of k integers with each of them divisible by k , then there exist integers $x_1, x_2, \dots, x_k$ and a positive integer r such that $a = \sum_{i=1}^k (k^r x_i)^2$ and $k \nmid x_i$ for some $1 \leq i \leq k$. Applying the lemma r times (the first time for the sum $\frac{a}{k^{2r-2}} = \sum_{i=1}^k (k x_i)^2$, etc.) we can find integers $y_1, y_2, \dots, y_k$ such that $a = \sum_{i=1}^k y_i^2$ and $k \nmid y_i$ for all $1 \leq i \leq k$.

Next we show that 8 is good. Let a be a positive integer which is the sum of squares of 8 integers with each of them divisible by 8. Then $64 \mid a$, hence $a \geq 64$ and $a = 1^2 + 4^2 + 4^2 + 4^2 + x_1^2 + x_2^2 + x_3^2 + x_4^2$ for some integers x_1, x_2, x_3, x_4 by Lagrange's four square theorem. Note that $x_1^2 + x_2^2 + x_3^2 + x_4^2 \equiv 7 \pmod{8}$ and the only way to get 7 as a sum of four quadratic residues modulo 8 is $1 + 1 + 1 + 4$. Therefore, $8 \nmid x_i$ for all $1 \leq i \leq 4$.

From this, we see that all positive integers which are divisible by an odd prime or 8 are good. The only cases left are $k = 1, 2, 4$. Indeed, 4 is not good since the nice integer $32 = 4^2 + 4^2 + 0^2 + 0^2$ is not funny by direct checking. Therefore, 1 and 2 are also not good numbers. $\square$